# Jinmyeong Choi

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## Summary

Master's student in Data Science at Seoul National University researching time series forecasting, representation learning, and time series foundation models. My research spans robust multivariate forecasting under asynchrony and missingness, analysis of statistical properties encoded in foundation model representations, and representation based process control. I am interested in developing interpretable and reliable AI systems that support scientific understanding and decision making.

## Education

**Seoul National University**, Seoul, Republic of Korea — *Sep. 2024 – Expected Feb. 2027*
*Master of Data Science*

**Seoul National University**, Seoul, Republic of Korea — *Mar. 2018 – Aug. 2024*
*Bachelor of Arts in Economics; Minor in Data Science for Humanities*

## Research Interests

- Time Series Forecasting
- Representation Learning and Interpretability
- Time Series Foundation Models

## Publications

[1] **Choi, J.**, Jang, J., Lee, S., and Kim, T. "*When Time Series Foundation Models Fail Under Cross-Channel Dependence*." *ICLR 2027 Under Review.*
- Investigated how time series foundation models use cross-channel information through controlled synthetic experiments and real-world forecasting tasks.
- Identified underuse of predictive dependencies and sensitivity to redundant inputs, and examined how covariate uncertainty affects forecast reliability.

[2] Jang, J., **Choi, J.**, Lee, S., and Kim, T. "*One Model, Three Views: Input-Condition Integration for Probabilistic Time Series Foundation Models*." *ICLR 2027 Under Review.*
- Studied input-condition integration across three forecast views from a single frozen time series foundation model.
- Evaluated history-based quantile corrections for more accurate forecasts and recovery of delayed covariate-driven responses.

[3] **Choi, J.**, Jang, J., Lee, S., and Kim, T. "*Interweaving Marginals into Multivariate Sample Paths: Training-Free Dependence Construction for Probabilistic Time Series Foundation Models*." *NeurIPS 2026 Workshop on Foundation Models for Temporal Systems (FMTS)*, 2026.
- Studied training-free coupling of frozen time series foundation model marginals into multivariate forecast sample paths.
- Isolated the effect of dependence construction under fixed empirical marginals, showing improved temporal and cross-channel dependence diagnostics.

[4] **Choi, J.**, Kim, T., and Dubrawski, A. "*Towards Universal Representation-Based Process Control*." *60th Asilomar Conference on Signals, Systems, and Computers*, 2026.
- Proposed a representation based process control framework that tests whether a system remains consistent with an empirical reference regime using foundation model embeddings.
- Developed a calibrated consistency testing pipeline that accommodates stable stationary and cyclostationary regimes beyond fixed parametric tests.

[5] **Choi, J.**, Shook, B., and Dubrawski, A. "*Non-Stationarity in the Embedding Space of Time Series Foundation Models*." *1st ICLR Workshop on Time Series in the Age of Large Models*, 2026.
- Investigated how distributional shifts and temporal dependence are preserved or suppressed in the embedding spaces of time series foundation models.
- Analyzed learned representations to reveal how statistical properties of time series are encoded, entangled, or discarded.

[6] Jang, J., Park, H., **Choi, J.**, and Kim, T. "*Towards Robust Real-World Multivariate Time Series Forecasting: A Unified Framework for Dependency, Asynchrony, and Missingness*." *The Fourteenth International Conference on Learning Representations (ICLR)*, 2026.
- Introduced ChannelTokenFormer, which models cross channel dependencies, asynchronous sampling, and missing intervals without interpolation.
- Contributed to architecture design and implementation, ablation studies, and evaluation on benchmark and industrial datasets.

## Research Experience

**Remote Research Collaborator**, Robotics Institute, Carnegie Mellon University *Feb. 2026 – Present*

- Continue collaboration with Prof. Artur Dubrawski on representation learning and monitoring methods for time series foundation models.

**Research Intern**, Auton Lab, Carnegie Mellon University *Aug. 2025 – Feb. 2026*

- Led an independent project on the internal representations of large scale time series foundation models under the supervision of Prof. Artur Dubrawski.
- Conducted controlled experiments on non-stationarity in embedding space and developed representation based methods for time series monitoring and process control.

**Visiting Scholar**, Carnegie Mellon University *Aug. 2025 – Feb. 2026*

- Completed graduate coursework in Introduction to Deep Learning, Natural Language Processing, Large Scale Media Analysis, and Studio Project.

**Research Assistant to Prof. Jeong Min Lee**, Seoul National University *Jan. 2024 – Feb. 2024*

- Conducted data preprocessing and analysis in Python for an evaluation of the "Ansim Income" policy.
- Collaborated with a Seoul Welfare Foundation researcher to analyze the policy's effects on welfare outcomes.

**Research Intern**, LAAL Lab, Seoul National University *Aug. 2023 – Aug. 2024*

- Built a financial time series dataset for stock price forecasting, compared MLP based forecasting models, and presented the findings.

## Teaching and Leadership

**Teaching Assistant**, Seoul National University *Mar. 2026 – Present*
*F37.101 Basic Computing: First Adventures in Computing*

- Lead hands-on lab sessions and guide students through practical Python programming exercises.

**Tutor**, Seoul National University *Mar. 2023 – Dec. 2024*
*F37.101 Basic Computing: First Adventures in Computing*

- Provided individual guidance on fundamental Python programming concepts, syntax, and problem solving.

**DTM Supporters**, Seoul National University *Sep. 2021 – Dec. 2021*

- Assisted international students with administrative tasks and cultural adaptation.

**President**, 7th L&K Sports Business Forum *Mar. 2022 – Aug. 2022*

**Seminar Team Leader and 1st Block Leader**, 32nd SNU Model United Nations *Mar. 2019 – Dec. 2019*

## Academic Service

**Reviewer**, ICML Workshop on Foundation Models for Structured Data (FMSD) *2026*

**Reviewer**, NeurIPS Workshop on Foundation Models for Temporal Systems (FMTS) *2026*

## Skills and Languages

**Programming and ML:** Python, PyTorch, NumPy, pandas, scikit-learn, Git, C/C++

**Languages:** English – High Proficiency (TOEFL 100; TEPS 465; CEFR C1), Mandarin – Intermediate

## Military Service

**Sergeant**, Republic of Korea Army; Honorably Discharged *Jan. 2020 – Jul. 2021*

- Served as a squad leader and completed Airborne/Paratrooper training, demonstrating leadership, adaptability, and teamwork under pressure.